



Proactive Cyber Threat Forecasting

Dark web intelligence, explainable AI, and architectural comparison for long-range prediction

Building on Almahmoud (2024), this research integrates dark web signals, deep learning architectures, and XAI techniques to shift cybersecurity from reactive incident response to anticipatory defence.



Can we give defenders more lead time by listening to the underground internet before attacks surface publicly?

Why Reactive Cybersecurity Is Not Enough

Three critical gaps leave defenders exposed while threat actors operate unchallenged.

Lagging Data Sources

Public vulnerability databases catalogue threats **after** exploitation. Dark web forums discuss exploits weeks or months earlier, yet this signal goes untapped.

Black-Box Predictions

Sophisticated models forecast without explaining *why* thereby hindering analyst action, resource allocation, and regulatory compliance (e.g. DORA).

No Architectural Consensus

Graph Transformers, Vision Transformers, and agentic LLMs are all plausible, yet nobody has systematically compared them for cyber threat time-series.



Result: the Colonial Pipeline attack (2021) demonstrates cascading real-world consequences of this intelligence gap.



A Richer Intelligence Platform

The dataset was expanded from the 2024 baseline. It is now covering **13.5 years** across six perspectives on the threat landscape.

1,292

Temporal Nodes

+810% vs original 142

2,126

Features

+1,385% vs original 143

11

Data Sources

+120% vs original 5

Incidents

Hackmageddon + U.
Maryland

Underground

Dark web pricing + 67
attack categories

Discourse

RSS feeds, Bluesky

Awareness

Google Trends, 86 topics

Vulnerability

NVD/CVE disclosed weaknesses

Conflict

ACLED geopolitical context

Five Architectures Compared

First systematic comparison for cyber threat time-series forecasting, all of which are pretrained via transfer learning on finance, epidemiology, and terrorism data.

1

Bayesian MTGNN: Baseline

Graph-based relational modelling with probabilistic uncertainty. Benchmark from Almahmoud (2024).

2

Graph Transformer: Spatial

GPS-enabled architecture combining local message passing with global attention. First use in this domain.

3

Vision Transformer: Temporal

Treats threat time-series as image patches. Excels at long-range dependencies with Bayesian MC dropout.

4

Spatio-Temporal Ensemble: Novel

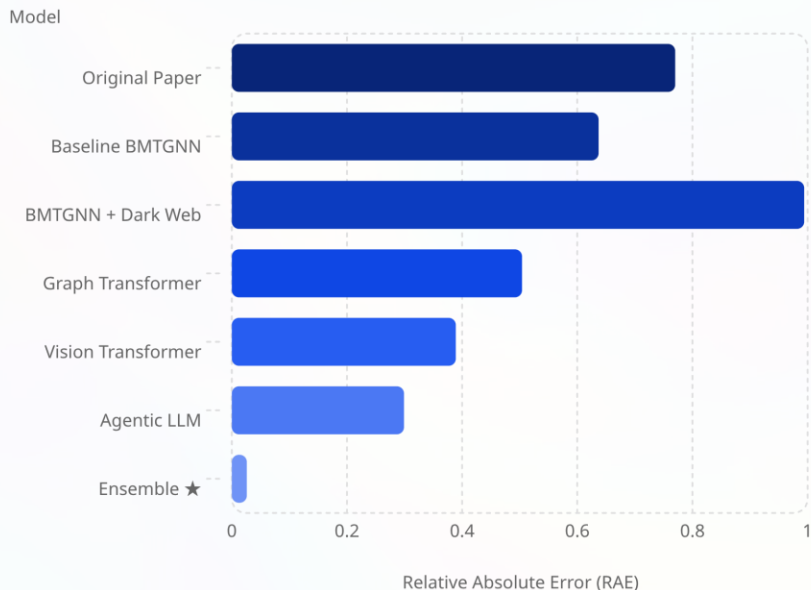
Original contribution: ViT + Graph Transformer fused via cross-attention and dynamic weighting.

5

Agentic LLM: Agentic

Seven-agent pipeline (Preprocess → ModelSelection → Analysis → XAI → Visualisation → PAT → Report). Hosted locally via Ollama + DeepSeek r1:8b.

Error Rates Across All Models

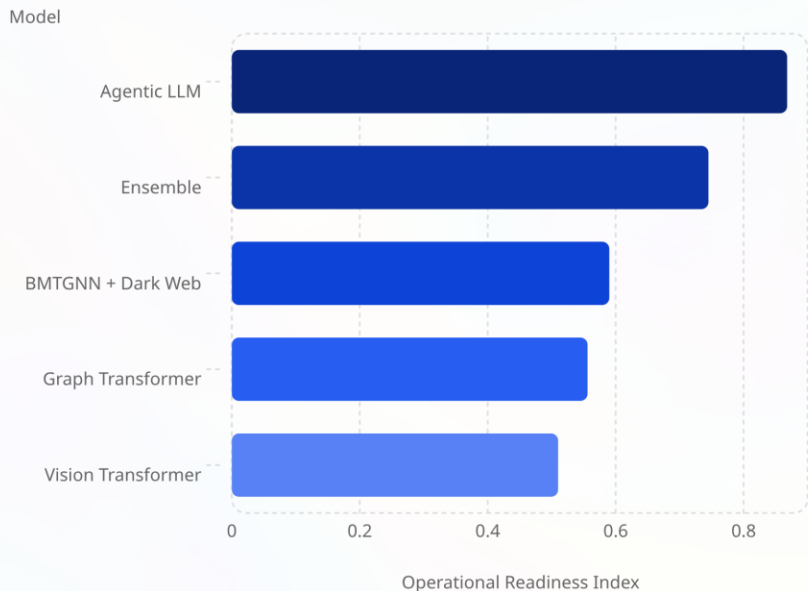


Lower is better (RAE)

The spatio-temporal ensemble achieves an **order-of-magnitude improvement** over all competing models (RAE 0.0258, RRSE 0.058) which is approximately **30× lower** than the original paper.

- 📄 The BMTGNN's deterioration with dark web data (RAE 0.994) suggests graph capacity does not scale linearly with feature dimensionality which is a critical architectural insight.

Operational Readiness vs Adversarial Robustness



Key trade-off

The ensemble achieves the **lowest error**, but the agentic LLM achieves the **highest operational readiness**. The right choice depends on whether the priority is raw accuracy or deployment practicality.

- ✔ **Adversarial robustness:** All dark-web-trained models scored ~0.926 which is dramatically higher than the baseline BMTGNN at 0.107. Exposure to naturally adversarial underground patterns hardens models against manipulation.



Making Predictions Actionable Through XAI

Four interpretability main methods applied across architectures, supporting DORA and SEC compliance with transparent, auditable predictions.



SHAP: Global Feature Importance

Driven by Google Trends, RSS feeds, and honeypot signals which is directionally consistent with known threat dynamics.



LIME: Local Explanations

Mid-window temporal context matters most as the model learns build-up patterns rather than reacting instantaneously.



MC Dropout: Uncertainty Quantification

Stable confidence intervals over time. High uncertainty flags novel patterns for human review; high confidence enables automated response.



DiCE + Causal: Counterfactuals & Causality

Sparse counterfactuals reveal targeted intervention points. Causal analysis separates true threat precursors from coincidental correlations.

Five Novel Contributions

Advancing theory and practice in proactive cybersecurity.

01

First Systematic Dark Web + Long-Term Forecast Integration

Temporally labelled corpus linking underground discussions to subsequent threat events which is a foundational research resource.

02

Comprehensive Architectural Comparison

Evidence-based guidance for selecting between B-MTGNN, Graph Transformer, Vision Transformer, and agentic LLM.

03

Novel Spatio-Temporal Ensemble Architecture

RAE 0.0258 (best in class). Cross-attention fusion of ViT and Graph Transformer with dynamic weighting and MC dropout.

04

Bayesian Uncertainty for Graph and Vision Transformers

First application in this domain. Guides resource allocation and distinguishes actionable intelligence from ambiguous signals.

05

First Specialised Agentic LLM for Cyber Threat Forecasting

Closes the Forecast → Explain → Recommend → Act loop in one workflow. Strongest ORI (0.868) among deployment-ready models.

Limitations & Future Directions

Current Limitations

- Dark web data limited to 2020–2024 due to structural complexity of scraping
- BMTGNN performance degraded with high-dimensional dark web features which presents architectural mismatch to investigate
- Agentic LLM susceptible to hallucination under adversarial conditions despite high ORI
- XAI scope reduced to fit project timeframe, the full causal analysis remains incomplete

Future Directions

- Extend dark web coverage pre-2020 with alternative archival sources
- Investigate hybrid BMTGNN variants for high-dimensional feature sets
- Validate ensemble in live SOC environment with real-time threat feeds
- Extend XAI to full causal inference and regulatory audit trail generation

Shifting From Reactive to Proactive

0.026

Best Error Rate

Ensemble RAE — 30× lower than original paper

0.868

Best Operational Readiness

Agentic LLM ORI

0.926

Adversarial Robustness

All dark-web-trained models

The spatio-temporal ensemble delivers best-in-class accuracy; the agentic LLM delivers best-in-class operational suitability. XAI transforms predictions into auditable, actionable intelligence.

Every hour of extended lead time is an hour defenders can use to patch, isolate, and prepare. This research is ultimately about expanding the horizon of preventable harm.

 Dataset to be released as a research resource (subject to appropriate access controls)



Potential Q&As

Why did the BMTGNN get worse with more data?

My hypothesis is that the graph learning components, specifically the adaptive graph construction, become unstable when the feature space expands dramatically. The model tries to learn too many relationships at once and the graph structure degrades. It's a known challenge in graph neural networks at scale, and it's one reason the transformer architectures, which don't have the same hard graph construction step, handled the expansion more gracefully.

Is the dark web data ethically obtained?

The data is publicly accessible on .onion sites: the same information is visible to any threat intelligence professional using Tor. I collected aggregated pricing and frequency data, not personal information or credentials. The collection methodology uses Tor Manager to ensure anonymity and does not interact with any marketplace in a way that would constitute a transaction or unlawful access.

How do you prevent the forecasting system from being gamed?

This is exactly what the adversarial robustness evaluation addresses. The dark-web-trained models scored around 0.926, meaning adversarial perturbations have very low success rates. The counterfactual analysis also helps here: it reveals which feature regions the model is most sensitive to, so we can harden those specifically. That said, it's an arms race, and ongoing monitoring for data drift via

Why not just use the ensemble in all cases?

Compute cost and operational overhead. The ensemble is excellent on accuracy but requires significantly more inference time and resources than the agentic LLM. In a production SOC environment, where latency and resource constraints are real, the agentic LLM's 0.87 ORI with 0.87 adversarial robustness may be the more pragmatic deployment choice. Architecture selection should always be driven by operational context.

How do the 3-year forecasts actually get used?

The gap analysis is the operational output. It identifies periods where a threat trajectory is projected to outpace the availability of defensive technologies, flagging windows of heightened vulnerability. Security teams can use these to prioritise R&D investment, procurement planning, and staffing decisions well in advance. The XAI outputs make these recommendations defensible to stakeholders and auditors.

Original Study Citation

- Almahmoud, Zaid & Yoo, Paul D. & Damiani, Ernesto & Choo, Kim-Kwang Raymond & Yeun, Chan Yeob, 2025. "Forecasting Cyber Threats and Pertinent Mitigation Technologies," Technological Forecasting and Social Change, Elsevier, vol. 210(C)
- <https://github.com/Birkbeck/msc-project-source-code-files-24-25-bgsbgsbgs42>